

MILESTONE 5: Deployment

In Milestone 5, the focus is on deploying the LegalEase application locally using FastAPI as the backend and Streamlit as the frontend. This milestone involves setting up a virtual environment, configuring dependencies, and verifying that all modules—from the Gemini API integration to document generation and downloads—work seamlessly in a real-world local environment. This ensures your system is production-ready and paves the way for cloud deployment in future stages.

Activity 5.1: Preparing the Application for Local Deployment

Step 1: Set Up a Virtual Environment

To ensure environment isolation and consistent dependency management, create and activate a virtual environment:

Step 2: Install Dependencies

Ensure that the backend (FastAPI) and frontend (Streamlit) dependencies are installed from a requirements.txt:

```
pip install -r requirements.txt
```

Step 3: Configure Environment Variables

Create a .env file to securely store your Gemini API key

Activity 5.2: Testing and Verifying Local Deployment

Step 1: Launch the FastAPI Backend

Start the backend server using Uvicorn:

Step 2: Launch the Streamlit Frontend

Run the app.py file to start the LegalEase UI:

Step 3: Test All Functionalities

Input document type, parties, terms, and date.

Press Generate Document to get AI-generated legal content from Gemini.

Use built-in options to:

- o Preview in dark mode HTML
- o Edit the text
- o Download in .txt, .docx, and .pdf formats

Check for:

- o Proper layout with logo and footer
- o Clean formatting of bullet points
- o Correct embedding of terms in the document